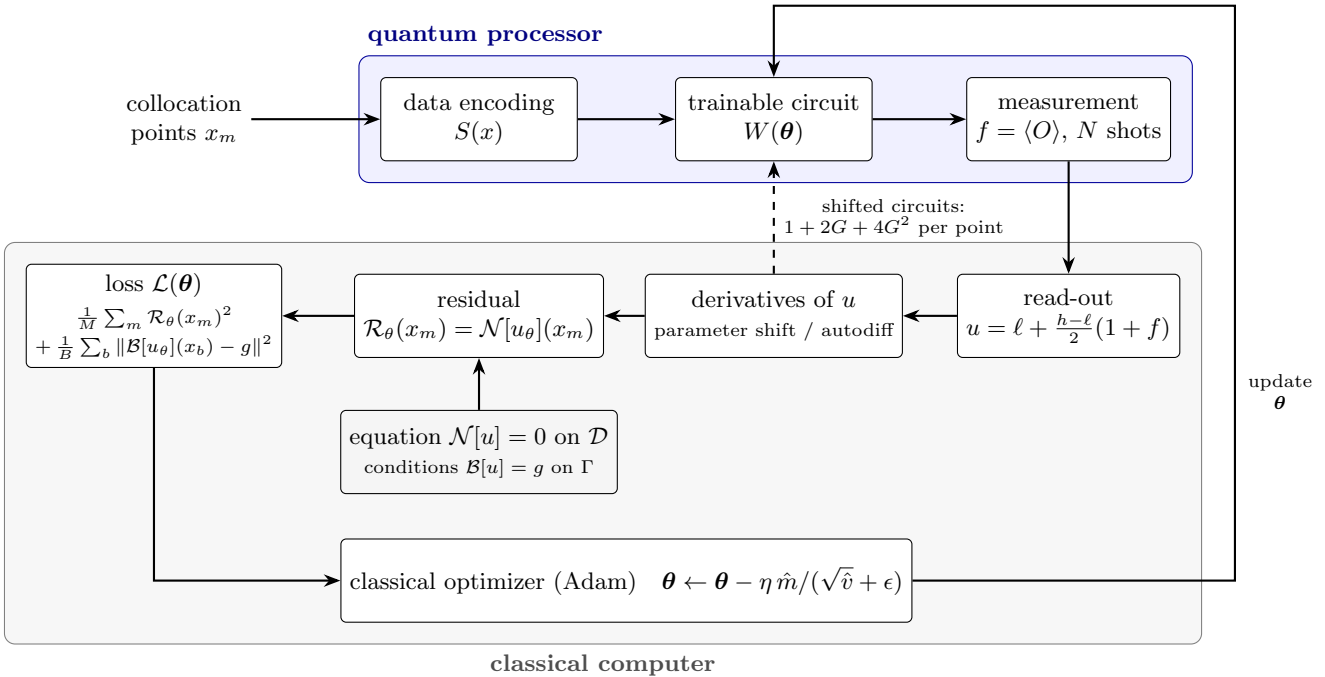

quantum processor
collocation points x_m
data encoding $S(x)$
trainable circuit $W(\boldsymbol{\theta})$
measurement $f = \langle O \rangle$, N shots
shifted circuits: $1 + 2G + 4G^2$ per point
read-out $u = \ell + \frac{h-\ell}{2}(1+f)$
derivatives of u parameter shift / autodiff
residual $\mathcal{R}_\theta(x_m) = \mathcal{N}[u_\theta](x_m)$
loss $\mathcal{L}(\boldsymbol{\theta})$ $\frac{1}{M}\sum_m \mathcal{R}_\theta(x_m)^2 + \frac{1}{B}\sum_b \|\mathcal{B}[u_\theta](x_b) - g\|^2$
equation $\mathcal{N}[u] = 0$ on $\mathcal{D}$ conditions $\mathcal{B}[u] = g$ on Γ
classical optimizer (Adam) $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\,\hat{m}/(\sqrt{\hat{v}} + \epsilon)$
update $\boldsymbol{\theta}$
classical computer